



Statistics of Financial Markets

Chapter 9: Value at Risk and Expected Shortfall



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Outline

Motivation

VaR Definition

VaR Estimation Methods

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Backtesting

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References

Motivation

- ▣ Quantifying downside risk in financial portfolios
- ▣ Regulatory requirements: Basel Accords and capital adequacy
- ▣ From volatility to tail risk measures

VaR Definition

- ▣ Formal definition: $\text{VaR}_\alpha = -\inf\{x : P(X \leq x) > \alpha\}$
- ▣ Interpretation: maximum loss at confidence level $1 - \alpha$ over a given horizon
- ▣ Properties: not subadditive in general
- ▣ Regulatory VaR: 99% confidence, 10-day horizon

Historical Simulation

- Non-parametric approach using empirical return distribution
- Rolling window estimation
- Advantages: no distributional assumptions
- Limitations: ghost effects, slow adaptation

Parametric VaR

- ▣ Variance-covariance method under normality
- ▣ $\text{VaR}_\alpha = \mu + \sigma \cdot z_\alpha$
- ▣ GARCH-based conditional VaR
- ▣ Student- t and skewed distributions for heavy tails

Monte Carlo VaR

- ▣ Simulation-based approach for complex portfolios
- ▣ Generating scenarios from fitted models
- ▣ Advantages: flexibility for non-linear exposures
- ▣ Computational considerations

Expected Shortfall

- ▣ Definition: $ES_\alpha = \mathbb{E}[X \mid X \leq -\text{VaR}_\alpha]$
- ▣ Coherent risk measure (subadditivity property)
- ▣ Relation to VaR: $ES \geq \text{VaR}$
- ▣ Basel III shift from VaR to Expected Shortfall

Backtesting

- ▣ Kupiec's (1995) proportion of failures (POF) test
- ▣ Christoffersen's (1998) conditional coverage test
- ▣ Independence of VaR violations
- ▣ Traffic light approach (Basel framework)

Applications

- ▣ VaR and ES computation for equity portfolios
- ▣ Comparing estimation methods on real data
- ▣ Backtesting results and model ranking
- ▣ Python implementation

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Conclusions

- ▣ Summary of VaR and Expected Shortfall methodologies
- ▣ Practical guidelines for risk managers
- ▣ Regulatory implications and current trends

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